**PUNJAB TIANJIN UNIVERSITY OF TECHNOLOGY (PTUT), LAHORE**
Department of Software Engineering Technology
**Course: Technical Report Writing | Class Assignment (10 Marks)**

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**FINAL PROJECT TOPIC**

# BugBountyTrack: A Secure Website for Reporting and Verifying Software Security Bugs

## 1. Statement of the Problem

When independent security testers find a weakness or bug in a company's website, they need a safe way to inform the company so it can be fixed. Today, most small businesses do not have a proper system for receiving these alerts. Instead, people have to send regular emails, fill out general 'Contact Us' forms, or message the company on social media. This causes two major problems:

- **Risk of Information Leaks:** The message explaining how the security flaw works is sent over normal email or sits in a shared office inbox. If the wrong person sees that email, they can use those exact instructions to hack the website before the company fixes it.
- **Closing Reports Without Real Proof:** In regular task-tracking tools, a programmer can simply click a button to mark a bug as 'Fixed' without showing any proof. No one checks if the code was actually repaired or tests the website again. Because of this, many serious security problems are marked as 'Solved' on paper while the website remains unsafe in reality.

If this problem is not solved, companies will falsely believe their websites are safe, and customer data will remain at risk of being stolen.

## 2. Aim and Objectives

### *Aim*

To design and build an easy-to-use website where people can safely report software security bugs to companies, and where companies must prove and test their repairs before closing a report.

### *Objectives*

1. **Gather User Needs:** Study what security reporters and company staff need so the website is simple and clear for everyone to use.
2. **Build a Safe Reporting Form:** Create a web page that automatically locks (scrambles) the bug report inside the user's browser before sending it, keeping the secret details completely private.
3. **Add a Bug Priority Calculator:** Build a simple rating tool that scores each bug from Low to Critical so companies know which problems to fix first.
4. **Create a Proof-of-Fix System:** Set up a step-by-step tracking process where programmers must attach proof of their code repair and pass a second check (re-test) before a bug report can be closed.
5. **Test the Website:** Test the finished system with sample bug reports to make sure it is easy to use, keeps information private, and prevents unfixed bugs from being closed early.

## 3. Proposed Solution

The proposed solution is BugBountyTrack, a central website that connects independent security testers with company software teams. What makes this project different from ordinary task-tracking apps are two main features:

- **Automatic Privacy Lock:** As soon as a reporter types the details of a bug, the website locks the message directly on their computer screen before sending it over the internet. Only the reporter and the company's authorized staff have the digital key to unlock and read it. Even if someone steals the website's main database, they will only see unreadable scrambled letters.
- **Verified Bug Closing:** Unlike normal apps where anyone can click 'Done,' BugBountyTrack requires real proof. A programmer must link their actual code update to the report. Then, the system moves the report to a 'Waiting for Re-test' stage. The report only closes after the original reporter or a team tester checks the website again and confirms the problem is truly gone.

## 4. Scope and Limitations

***Project Scope***

- **Included Features:** A public page showing the company's security rules, bug submission forms for both guests and registered users, automatic message locking, a bug priority calculator, a step-by-step fix checker, and a permanent history log of all actions.
- **Platform:** Works on any standard computer or laptop web browser (such as Google Chrome, Firefox, or Microsoft Edge).
- **Target Users:** Independent security testers, small-to-medium software companies, and university project teams.

***Project Limitations***

- **No Automatic Hacking:** The website is a reporting and management tool for humans; it does not automatically scan or attack websites on its own.
- **No Online Money Transfers:** The system tracks bug reports and repairs, but it does not handle bank accounts or send cash rewards.
- **Internet Required:** Users must have an active internet connection to submit and check reports.

## 5. Significance of the Study

This project provides practical benefits to three main groups:

- **Small Businesses and Startups:** Gives smaller companies a professional, affordable way to receive security warnings without paying thousands of dollars for expensive security agencies.
- **Security Testers and Students:** Gives helpful researchers a safe, legal, and private place to report problems and get fair credit for helping companies improve.
- **Everyday Internet Users:** Because the system forces companies to actually test and prove their repairs, the websites people use every day become much safer, protecting personal passwords and private information from hackers.

## 6. Organization of the Document (Outline of Chapters)

The rest of this project report is divided into five clear chapters:

**Chapter 1: Introduction —** Explains the problem being solved, the main aim and step-by-step objectives, what the project covers, and who benefits from it.

**Chapter 2: Literature Review —** Looks at how companies currently handle bug reports, points out the weaknesses in existing tools, and explains why a better system is needed.

**Chapter 3: System Analysis and Design —** Shows the planning behind the project, including user needs, how the system is organized, database tables, and step-by-step flowcharts.

**Chapter 4: Implementation and Testing —** Describes how the website was built, shows the finished screens, and shares the results of testing the system with sample bug reports.

**Chapter 5: Conclusion and Future Work —** Summarizes what the project achieved, lessons learned during development, and ideas for adding new features in the future.